Computer Vision in Archaeology Training School

Institute of Archaeology of the Czech Academy of Sciences, Brno (ARÚB)

14–18 September 2026

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My name is Hüseyin Erdoğan, and I am a PhD student in Digital Archaeology at the Department of History and Cultures, University of Bologna. My research combines digital archaeological documentation with standardised workflows aligned with the FAIR principles and supported by Semantic Web technologies. My research focuses on the process-oriented documentation of High-Density Survey and Measurement techniques, as well as the management of archaeological excavation documentation systems.

I am working on the digitisation of the rock art figures at Rupe Magna, located in the Valtellina region of northern Italy, using computational imaging techniques. Furthermore, by working on the digital documentation processes of the Ca' Nova archaeological excavation at Albareto in Parma and the semantic modelling of the data produced during these processes, I am interested in the structuring of archaeological knowledge and the enhancement of its reusability. In particular, my research background in the fields of computer vision in archaeological analysis and the production of archaeological knowledge ensures that this training programme is closely aligned with my research interests.

I decided to take part in the 'Computer Vision in Archaeology Training School', which I saw as a valuable opportunity to further develop my academic skills in fields closely related to my current research. The training school took place in Brno from 14 to 18 September 2026, and this was my first visit to the city. It was drizzling when I arrived in Brno, but the city had already given me a warm welcome. I went to my accommodation and got ready for the first day of the school.

Day 1 – 14 September 2026

The Institute building made a deep impression on me at first glance. The team members were extremely kind and welcoming; this was the first indication that the programme would run smoothly and that keeping up with the day's activities would be both organised and stress-free.

The instructors included *Petr Pajdla, Ronald Harasim* and *Filip Hájek* from the *Institute of Archaeology of the Czech Academy of Sciences, Brno (ARÚB)*; *Anastasia Eleftheriadou* from *Ludwig-Maximilians-Universität München*; and *Ashley Green* from the *University of Gothenburg*. *Zuzana Kopáčová, Vít Kozák, David Spáčil* and *Tereza Siková* also contributed to the organisation and support of the school.

In our first session, led by *Petr Pajdla*, we were introduced to computer vision and its applications in archaeology. We formed small groups with those sitting next to us; this encouraged our first interactions as participants in the course. Participants from different countries and cultural backgrounds were working on various types of cultural heritage and research topics, and we shared many common interests. This enabled us to discuss the questions raised during the session and to discover other archaeological applications as part of the learning process. In this session, the distinctions that would form the basis for the rest of the week were introduced: classification determines what is depicted in an image; object detection determines what is present and where it is located; segmentation delineates regions at the level of individual pixels; while keypoint detection identifies the locations of specific reference points. These concepts provided a framework for understanding the various case studies discussed later.

A concrete example was the Digital Archive of the Archaeological Map of the Czech Republic, an infrastructure to which team members contributed. This archive provides access to archaeological documents, fieldwork records, excavation sites, individual finds and a library of 3D models. By bringing together photographs, texts, maps, plans and other digital resources, this archive demonstrates how digital infrastructures can preserve archaeological knowledge and make it accessible to researchers and a wider audience. This has provided a useful context for assessing how computer vision technology can support work carried out with archaeological collections.

We then moved on to an introductory Python session led by *Filip Hájek*. For me, this formed the foundation of the entire programme. Before we started writing code, we need to understand what we want to achieve and how to express this task in instructions that a computer can follow. Although this session was introductory in nature, it helped me understand the logic behind the practical exercises we would be carrying out throughout the week.

> *The ice-breaker held on the first evening was very well organised and gave us the chance to get to know one another. Everyone I met had brought different ideas and interesting projects with them; it was this opportunity that had brought us all together in one place. That evening, I left with the feeling that many of these participants could play a significant role in shaping the future of digital archaeology.*

Day 2 – 15 September 2026

The second day was quite busy right from the start. Working alongside *Petr Pajdla*, we examined a computer vision model on Google Colab and moved on to practical aspects of image labelling. I'd been wanting to experiment with labelling tools for a while but hadn't had the chance. I certainly wasn't alone: according to data shared during the session, around 80 per cent of the group had never used an image labelling tool before.

Working with CVAT and SAM 2 helped me understand just how important labelling is in the research process. If we wish to obtain scientifically meaningful results from image segmentation, we must take particular care with the examples we label and use to train or evaluate a model. Model-assisted tools can help to determine the boundaries of objects, but the researcher must still assess those boundaries and decide what the labels signify. This study clearly demonstrated to me that labelling involves not only technical work but also a process of archaeological judgement.

Furthermore, we examined how archaeological records and images are organised via the Archaeological Map of the Czech Republic. Together with the use of structured categories, this provided a concrete example of the importance of organising archaeological information in a consistent manner.

The artwork dataset contained approximately ten thousand photographs and annotations covering 112 types of artwork. We examined how this information was represented in the COCO format, using JSON to organise the dataset's details, licences, images, categories and annotations. We also compared this with YOLO annotation formats, which typically use a text file where a separate line is allocated for each object annotated in each image, and with Pascal VOC, which uses XML files.

The practical session provided the answer to a question I'd brought with me to the course: what needs to be done before training a model on archaeological images? When it comes to supervised workflows, a significant part of the answer lies in preparing and verifying the labels ourselves. Labelling archaeological artefacts using

the school's CVAT setup helped me understand how human interpretation becomes an integral part of the data on which the model is based. This was one of the most valuable lessons of the day.

Day 3 – 16 September 2026

The third day began with a workshop on microscopy datasets led by *Anastasia Eleftheriadou*. I found her presentation particularly clear and useful. As someone who works with images, it was invaluable for me to revisit the fundamental components of a digital image: pixels, RGB colour values and histograms, which became much more meaningful when considered in relation to the analysis we wished to carry out.

Before moving on to the practical exercises, there was also a useful opportunity to ask questions. We worked with the provided datasets and Google Colab notebooks; I also created my own copies to start experimenting with the methods. The practical session covered image pre-processing, thresholding, edge detection, connected component analysis and the extraction of regions of interest. Techniques such as Canny edge detection helped to demonstrate how image processing can reveal or isolate specific structures.

The exercises on archaeological micromorphology were particularly interesting to me. Micromorphological analysis examines the components, structures and textures visible in thin sections of sediment. Computer vision can support this work by helping to visually distinguish different regions within an image and define their boundaries.

During an archaeological excavation I took part in a short while ago, we had collected samples from a stratigraphic section for micromorphological analysis. This session was therefore directly linked to my experience in the field. This session helped me understand the next stage of the process: how the images obtained from these samples could be examined digitally to support archaeological interpretation. Seeing this connection between the sample collection process during the excavation and the subsequent image analysis made the workshop particularly meaningful for me.

> *The social event that evening had been carefully organised, and a wide variety of food was on offer. By this point, the institute had really started to feel like home to us. Although we had attended an introductory event on Monday, it felt as though it was only that evening that we truly began to feel at ease with one another. This event gave us the opportunity to socialise, exchange ideas and discuss our shared interests more freely.*

Day 4 – 17 September 2026

The fourth day began with a session on satellite imagery led by *Filip Hájek*. We worked on Sentinel-2 multispectral imagery of Cornești-Iarcuri, a large fortified settlement dating from the Bronze Age, located in Romania. This exercise demonstrated how vegetation growing over buried ditches and walls can affect the imagery, and how field marks visible in satellite imagery come into being.

Using a Google Colab notebook, we analysed spectral bands, spectral signatures and the Normalised Difference Vegetation Index (NDVI). The workflow involved reading georeferenced raster data, creating true-colour and false-colour composites, and enhancing crop signals using local background subtraction and the Frangi ridge filter. We then introduced a Random Forest classifier trained on manually selected points. False positives from

the initial model could be added as ground-truth negatives during retraining before the results were converted into polygons for use in GIS.

This case study helped me understand how the various stages of image processing and classification can be integrated into an archaeological workflow. It also highlighted why it is important to verify the results: identifying a pattern in the vegetation cover is a step towards interpretation, and its archaeological significance still needs to be assessed.

In the afternoon, *Ashley Green* presented a case study on rock art and discussed Scandinavian rock art, spanning the period from the Stone Age to the Iron Age, focusing on both figurative and non-figurative imagery. We also worked on a Google Colab notebook as part of a model training exercise and examined the different tasks addressed by classification, object detection and segmentation models. This session was particularly important to me because I am currently working on rock art datasets from Rupe Magna in the Valtellina region of northern Italy. Having the opportunity to explore the field of computer vision using a different rock art dataset made me realise that these methods are directly relevant to my own research. I was delighted that the programme included this topic, and it has strengthened my desire to explore these approaches in greater depth within the context of Rupe Magna.

Day 5 – 18 September 2026

The fifth and final day began with another interesting case study presented by *Ronald Harasim*: a coin dataset. The session brought together advanced topics and a combination of different methods, including large language models and visual-language models. As well as learning what these technologies are capable of, we also need to understand how how their outputs are generated, what their limitations are, and how they can be assessed within the context of archaeological research.

Looking back, it was impossible not to notice from the very first day just how exceptionally well-organised the school was. For the first time, I had the opportunity to spend a week within a community whose members shared many of my interests and goals. The practical sessions were invaluable, but the conversations that developed throughout the programme and the sense of belonging were just as valuable. I would gladly take part again for a similar experience.

Computer vision and machine learning are becoming increasingly important for archaeology, and I believe that we need both technical expertise and methodological rigour when using these tools. The value of these tools lies in their potential to expand and enrich archaeological knowledge. For me, this training programme has reinforced the contribution of these methods to archaeological knowledge production and highlighted the importance of considering its epistemological dimensions: how data are prepared, how interpretative decisions are made, and how evidence generated through computational methods is evaluated. This is because these processes influence not only the scope of the knowledge produced, but also how it is grounded, to what extent it can be considered reliable, and how it shapes our understanding of archaeological reality.

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